UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

STAT 576: Empirical Process Theory and Weak Convergence Fall 2025
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Lecture 22

Content.

1. Sudakov-Fernique
2. Sudakov Minoration
3. Operator Norms of Random Matrices

1 Sudakov-Fernique

Lemma 1. *Suppose $X \in \mathbb{R}^N$ and $Y \in \mathbb{R}^N$ satisfy the condition of Slepian's Lemma. Then for $\forall \tau \in \mathbb{R}$,*

$$\mathbb{P}\left[\max_i X_i \geq \tau\right] \leq \mathbb{P}\left[\max_i Y_i \geq \tau\right]$$

Remark.

1. *Directly implies that* $\mathbb{E}[\max_i X_i] \leq \mathbb{E}[\max_i Y_i]$
2. Proof Idea *Let $h(x)$ be a smooth approximation to $1_{(-\infty,\tau)}$. Define*

$$F(x) = h(x_1) \cdots h(x_n)$$

$$\frac{\partial^2 F}{\partial x_i \partial x_j} = h'(x_i)h'(x_j) \prod_{k \neq i,j} h(x_k) \geq 0.$$

Thus, $\mathbb{E}[F(X)] \geq \mathbb{E}[F(Y)]$. Take limits using DCT.

Sometimes, it's useful to compare two processes using the pseudometric.

$$P_X(i,j)^2 = \mathbb{E}[(X_i - X_j)^2]$$

Theorem 1 (Sudakov-Fernique)**.** *Let* $(X_t)_{y\in T}$ *and* $(Y_t)_{t\in T}$ *be two* Gausisan processes *with zero mean and that*

$$\mathbb{E}[(X_t - X_s)^2] \leq \mathbb{E}[(Y_t - Y_s)^2], \quad \forall t, s \in T.$$

Then:

$$\mathbb{E}\sup_{t\in T} X_t \leq \mathbb{E}\sup_{t\in T} Y_t$$

2 Gaussian Contraction

Suppose $\phi_1, \cdots \phi_d : \mathbb{R} \to \mathbb{R}$ with $\phi_i(0) = 0$ such that ϕ_i is 1-Lipschitz

$$|\phi_i(x) - \phi_i(y)| \leq |x - y|, \, \forall x, y \in \mathbb{R}.$$

For all $\theta \in \mathbb{R}^d$ we define

$$\phi(\theta) := (\phi_1(\theta_1), \phi_2(\theta_2), \cdots, \phi_d(\theta_d)) \in \mathbb{R}^d.$$

For any $T \subset \mathbb{R}^d$,

$$\phi(T) := \{\phi(\theta) : \theta \in \mathbb{R}^d\}.$$

Theorem 2.

$$\mathbb{E}\sup_{\theta\in T}\langle g, \phi(\theta)\rangle \leq \mathbb{E}\sup_{\theta\in T}\langle g, \theta\rangle$$

where $g \sim \mathcal{N}(0, I_d)$*. In particular,*

$$\mathcal{G}(\phi(T)) \leq \mathcal{G}(T).$$

Proof.

$$X_\theta := \langle g, \phi(\theta)\rangle$$
$$Y_\theta := \langle g, \theta\rangle$$

where $\mathbb{E}[X_\theta] = \mathbb{E}[Y_\theta]$.

$$\begin{aligned}
\mathbb{E}[(X_\theta - X_{\theta'})^2] &= \mathbb{E}[\langle g, \phi(\theta) - \phi(\theta')\rangle^2] && (1)\\
&= \|\phi(\theta) - \phi(\theta')\|_2^2 && (2)\\
&= \sum_{j=1}^{d} \left|\phi_j(\theta_j) - \phi_j(\theta'_j)\right|^2 && (3)\\
&\leq \sum_{j=1}^{d} |\theta_j - \theta'_j|^2 = \|\theta - \theta'\|_2^2, \text{ using 1-Lipschitz} && (4)\\
&= \mathbb{E}[(Y_\theta - Y_{\theta'})^2]. && (5)
\end{aligned}$$

The result now follows from Sudakov-Fernique. □

3 Sudakov Minonartion

So far: we upper bound supremum of Sub-Gaussian processes.

1. One-step discretization
2. Chaining/Dudley entropy

Now: Lower bound for Gaussian processes.

Theorem 3 (Sudakov Minoration)**.** *Let $\{X_\theta\}_{\theta\in T}$ be a centered Gaussian process on T. Then;*

$$\mathbb{E}\left[\sup_{\theta\in T} X_\theta\right] \geq \sup_{\delta>0}\left\{\frac{\delta}{2}\sqrt{\log M_X(T,d,\delta)}\right\},$$

*where $M_X(T,d,\delta)$ is the δ-packing number of T in the metric (*pseudometric*)*

$$d(\theta,\theta') = \sqrt{\mathbb{E}\left[(X_\theta - X_{\theta'})^2\right]}$$

Proof. Take a δ-packing set $\{\theta^1,\cdots,\theta^M\}\subset T$. for T, $d(\theta^i,\theta^j)\geq\delta$, $M = M_X(T,d,\delta)$.

$$\mathbb{E}\left[\sup_{\theta\in T} X_\theta\right] \geq \mathbb{E}\left[\max_{1\leq i\leq M} X_{\theta_i}\right]$$

$$Y_i := \frac{\delta}{\sqrt{2}} Z_i, \text{ where } Z_i \overset{iid}{\sim} \mathcal{N}(0,1)$$

$$\mathbb{E}\left[(X_{\theta_i} - X_{\theta_j})^2\right] = d(\theta_i-\theta_j)^2 \geq \delta^2$$

$$\mathbb{E}\left[(Y_i - Y_j)^2\right] = \mathbb{E}\left[\left(\frac{\delta}{\sqrt{2}}(Z_i - Z_j)\right)^2\right] = \delta^2.$$

By Sudakov-Fernique ($\mathbb{E}\left[\max_{1\leq i\leq m} Z_i\right] \geq (1-e^{-1})\sqrt{2\log M}$),

$$\mathbb{E}\left[\max_{i\leq M} X_{\theta_i}\right] \geq \mathbb{E}\left[\max_{1\leq i\leq m} Z_i\right]\frac{\delta}{\sqrt{2}} \geq \frac{\delta}{2}\sqrt{\log M}$$

□

Example : $\mathcal{G}(B_2^d) = \mathbb{E}\left[\sup_{\theta\in B_2^d}\langle\theta, g\rangle\right]$,

$$B_2^d = \left\{\theta\in\mathbb{R}^d : \|\theta\|_2 = 1\right\}.$$

If $X_\theta := \langle \theta, g \rangle$, $d(\theta, \theta') = \|\theta - \theta'\|$. So, Sudakov's minoration gives:

$$\mathcal{G}(B_2^d) \geq \sup_{\delta > 0} \left\{ \frac{\delta}{2} \sqrt{\log M} \right\}$$

$$\log M \geq \log \mathcal{N}(B_2^d, \delta, \|\cdot\|) \geq \log \left(\frac{1}{\delta} \right)^d.$$

Thus,

$$\mathcal{G}(B_2^d) \geq \sup_{\delta > 0} \left\{ \frac{\delta}{2} \sqrt{d \log \frac{1}{\delta}} \right\} \asymp \Omega(\sqrt{d})$$

4 Application: Bounding Covering Numbers of $T \subset \mathbb{R}^n$

Corollary 1 (Vershynin)**.** *Let $T \subset \mathbb{R}^n$. Then; $\exists c > 0$ such that*

$$\mathbb{E}\left[\sup_{t \in T} \langle g, t \rangle \right] \geq c\varepsilon \sqrt{\log \mathcal{N}(T, \varepsilon, \|\cdot\|)}.$$

Theorem 4. *Let P be a polytope in $\mathbb{R}^n$ with N vertices, contained in unit Euclidean ball. Then $\exists \mathcal{C} > 0$ such that $\forall \varepsilon > 0$,*

$$\mathcal{N}(P, \varepsilon) \leq N^{\frac{\mathcal{C}}{\varepsilon^2}}. \qquad (*)$$

Remark. *Note that $*$ is independent of dimension n.*

Proof. Let $x_1, \cdots, x_N$ be the vertices of P. In particular, for any $x \in P$, $\exists \lambda_1, \cdots, \lambda_N \geq 0$, $\sum_i \lambda_i = 1$,

$$x = \sum_{i=1}^N \lambda_i x_i.$$

$$\mathbb{E}\left[\sup_{x \in P} \langle g, x \rangle \right] = \underbrace{\mathbb{E}\left[\max_{1 \leq i \leq N} \langle x_i, t \rangle \right]}_{x_i\text{'s contained in unit ball}} \leq \mathcal{C}' \sqrt{\log N}.$$

Thus:

$$\mathcal{C}' \sqrt{\log N} \geq c\varepsilon \sqrt{\log \mathcal{N}(P, \varepsilon)}$$

$$\left(\frac{\mathcal{C}'}{c} \right)^2 \frac{1}{\varepsilon^2} \log N \geq \log \mathcal{N}(P, \varepsilon) \Rightarrow \mathcal{N}(P, \varepsilon) \leq N^{\frac{\mathcal{C}}{\varepsilon^2}}, \text{ where } \mathcal{C} := \left(\frac{\mathcal{C}'}{c} \right)^2.$$

□

5 Operator Norms/Singular Values of Random Matrices

Setup. Let $A \in \mathbb{R}^{m\times n}$ with iid $\mathcal{N}(0,1)$ entries. We will bound $\sigma_{\min}(A), \sigma_{\max}(A)$.

Remark. *There are asymptotic results, $m, n \to \infty$ (Bai-Yin's law). Our goal is non asymptotic.*

$$\begin{aligned}\sigma_{\max}(A) = \|A\| &= \sqrt{\lambda_{\max}(A^TA)} \\ &:= \sup_{u\in\mathbb{S}^{n-1}} \|Au\|_2,\end{aligned}$$

where $\mathbb{S}^{n-1} = \{u \in \mathbb{R}^n : \|u\|_2 = 1\}$.

$$\sigma_{\min}(A) = \inf_{u\in\mathbb{S}^{n-1}} \|Au\|_2.$$

Theorem 5. *For $A \in \mathbb{R}^{m\times n}$, with iid $\mathcal{N}(0,1)$ entries,*

$$\sqrt{m} - \sqrt{n} \le E\sigma_{\min}(A) \le E\sigma_{\max}(A) \le \sqrt{m} + \sqrt{n}$$

This is a non-asymptotic variant of Bai-Yin law.

Theorem 6. *For any $t > 0$, and A as above,*

$$\sqrt{m} - \sqrt{n} - t \le \sigma_{\min}(A) \le \sigma_{\max}(A) \le \sqrt{m} + \sqrt{n} + t$$

with probability at least $1 - 2\exp(-\frac{t^2}{2})$.

Remark. *If $m \gg n$, then $\sigma_{\max}(A) \approx \sigma_{\min}(A) \approx \sqrt{m}$*

Proof of Theorem 5.

Lemma 2.

$$\|A\| = \sup_{u\in\mathbb{S}^{n-1}, v\in\mathbb{S}^{m-1}} \langle Au, v\rangle$$

Proof. We'll prove

$$\|Au\| = \sup_{v\in\mathbb{S}^{m-1}} \langle Au, v\rangle$$

$$\langle Au, v\rangle \le \|Au\| \cdot \|v\| = \|Au\|, \text{ by Cauchy-Schwarz}$$

as we can take $v = \frac{Au}{\|Au\|}$ we have

$$\langle Au, v\rangle = \langle Au, \frac{Au}{\|Au\|}\rangle = \frac{\|Au\|^2}{\|Au\|} = \|Au\|$$

□

Roadmap:

1. $X_{uv} := \langle Au, v \rangle$ for $(u, v) \in T = \mathbb{S}^{n-1} \times \mathbb{S}^{m-1}$
2. $\mathbb{E}[X_{uv}] = 0 \; \forall u, v \in T$

Next lecture:

$$\mathbb{E}\left[(X_{uv} - X_{u'v'})^2\right] \leq \|u - u'\|^2 + \|v - v'\|^2$$

where

$$\|u - u'\|^2 + \|v - v'\|^2 = \mathbb{E}\left[(Y_{uv} - Y_{u'v'})^2\right]$$

for

$$Y_{uv} = \langle g, u \rangle + \langle h, v \rangle$$

for $g \sim \mathcal{N}(0, I_n)$ and $h \sim \mathcal{N}(0, I_m)$. □